



DATA CLEANING & PREPARATION **REPORT**

Project 1 – Data Cleaning & Preparation

Submitted By

Name: Sofia Ahmed

Batch: 20 May – 20 June, 2026

Department: Data Analytics

Organization

Decode Labs

Dataset

E-Commerce Orders Dataset

Dataset Named as

Dataset for Data Analytics Project 1

Submission Date

5 June, 2026

PROJECT OVERVIEW

Objective

The objective of this project was to transform a raw e-commerce dataset into a clean, consistent, and analysis-ready dataset. The dataset contained data quality issues such as missing values, duplicate records, inconsistent text formatting, invalid date formats, and potential duplicate identifiers.

The cleaning process was performed using Python in Jupyter Notebook following standard data preparation practices commonly used in industry.

Tools and Technologies Used

- Python
 - Pandas
 - NumPy
 - Jupyter Notebook
-

Dataset Information

- Total Records: 1200
- Total Columns: 14

Columns Included:

- | | |
|-------------------|------------------|
| • OrderID | • PaymentMethod |
| • Date | • OrderStatus |
| • CustomerID | • TrackingNumber |
| • Product | • ItemsInCart |
| • Quantity | • CouponCode |
| • UnitPrice | • ReferralSource |
| • ShippingAddress | • TotalPrice |
-

DATA CLEANING CHANGE LOG

Change ID	Description	Impact	Status
CR001	Identified missing values across the dataset	Data quality issues detected for correction	Resolved
CR002	Missing numerical values were imputed using Median values	Preserved records without data loss	Resolved
CR003	Missing categorical values were imputed using Mode values	Maintained dataset completeness	Resolved
CR004	Date column converted to a standard date format (YYYY-MM-DD)	Improved consistency and usability	Resolved
CR005	Invalid date values were identified and corrected	Eliminated date formatting errors	Resolved
CR006	Leading and trailing spaces removed from text columns	Improved text consistency	Resolved
CR007	Text fields standardized into a consistent format	Reduced category duplication caused by formatting differences	Resolved
CR008	Duplicate rows identified and removed	Eliminated redundant records	Resolved
CR009	Duplicate Order IDs identified and corrected	Achieved unique record identification	Resolved
CR010	Numeric columns converted to appropriate numeric data types	Enabled accurate calculations and analysis	Resolved
CR011	TotalPrice values validated and recalculated using Quantity × UnitPrice	Improved financial data accuracy	Resolved
CR012	Final validation checks performed on the cleaned dataset	Confirmed compliance with project requirements	Resolved

FINAL DATA QUALITY VALIDATION

The cleaned dataset was subjected to a final validation process to ensure compliance with all project requirements.

Validation Results:

- Total Missing Values: 0
- Duplicate Rows: 0
- Duplicate IDs: 0
- Incorrect Dates: 0

Result: PASS

Final Quality Check Report

```
[65]: print("Total Missing Values:", df.isnull().sum().sum())
      print("Duplicate Rows:", df.duplicated().sum())
      print("Duplicate IDs:", df["OrderID"].duplicated().sum())

      date_check = pd.to_datetime(
          df["Date"],
          errors="coerce"
      ).isnull().sum()

      print("Incorrect Dates:", date_check)
```

```
Total Missing Values: 0
Duplicate Rows: 0
Duplicate IDs: 0
Incorrect Dates: 0
```

The dataset successfully met all quality assurance requirements and is ready for analytical and reporting purposes.

PROJECT OUTCOME

The raw dataset was successfully transformed into a clean and reliable dataset through systematic data preparation techniques. Missing values were handled using appropriate statistical methods, duplicate records were removed, identifier uniqueness was ensured, date formats were standardized, and all columns were validated for consistency.

The resulting dataset is suitable for business reporting, dashboard development, exploratory data analysis, and future machine learning applications.

CONCLUSION

Data cleaning is one of the most critical phases of the data analytics lifecycle because the quality of insights depends directly on the quality of the underlying data.

During this project, all identified data quality issues were successfully resolved while preserving the integrity of the dataset. The final dataset contains no missing values, no duplicate records, no duplicate identifiers, and no invalid dates. Through this process, a reliable and analysis-ready data source was produced that can support accurate business decision-making.

This project demonstrates the practical application of industrial data cleaning and preparation techniques using Python and Jupyter Notebook.
